1 **(a)** Define the term *moment of a force*.

.....

.....

..... [1]

(b) A uniform rigid rod of mass 30 kg is attached to a vertical wall by a hinge as shown in Fig. 1. The other end of the rod is held to the ceiling by a cable.

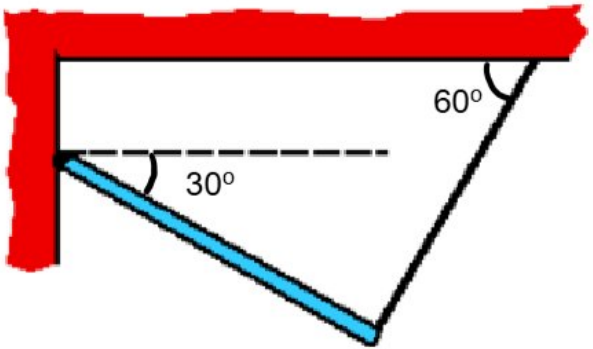


Fig. 1

(i) Draw and label clearly all the forces acting on the rod in Fig. 1 [2]

(ii) Show that the tension T in the cable is 127 N. [1]

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(iii) Determine the force acting on the rod by the hinge.

force = ... N

direction = ... [4]

[Total: 8]

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